

Title of dataset	Livestock numbers 1971–2017
Variables	geography_type: Geography breakdown geography_name: Geography location year: Year of survey/census animal: Livestock type count: Number of livestock
Environmental reporting topic	Resource use and management, and other human activities
Environmental reporting category	The accuracy of this data is of high quality
Environmental report	Environment Aotearoa 2019
Relevant measure on the Environmental Indicators, Te taiao Aotearoa website	Livestock numbers
Other data and reports which relate to this measure	Related indicators: - Agricultural and horticultural land use - Change in farm numbers and farm size - Change in use of Māori land for primary production Related content: - Our land 2018 - Agricultural production statistics: June 2017 (final)
Methodology (collection & analyses)	Every five years Stats NZ conducts a census of all farms (52,300 in 2017), and undertakes a sample survey in non-census years (30,400 farms were surveyed in 2016). Census years will not contain sampling error. However, there will be error with the imputation process applied to the data, given that not all farms respond. The population of the census and survey is all businesses registered for goods and services tax and classified by Stats NZ's Business Frame as being engaged in horticulture, cropping, livestock farming, or exotic forestry operations. Respondents report the number of livestock as at 30 June for each farm owned or leased. Here, we report only on cattle, sheep and deer, we do not report on other types of farmed animals such as pigs, chickens and goats. The source of the target population of farms has been based on the Business Frame from 2002 and Business Register from 2015. From the target population, survey years select samples stratified by region, size, and farm type.

	We present livestock numbers data back to 1971 at the New Zealand level, and back to 1990 at the region and island level. Territorial authority (TA) and grid level data is presented for census years back to 1994. We present past data as it relates to current boundary names and areas, with the exception of the pre-1993 Nelson-Marlborough Region, which has been left as is.
Limitations to data & analysis	We advise caution when interpreting long-term trends as the census and survey changed their questionnaire design and farm coverage between 1994 and 2002, and there were breaks in time series. No agricultural survey was conducted in 1997, 1998, 2000, and 2001. Between 1994 and 1996, the census and survey population included businesses registered for GST and recorded on the Business Frame as engaged in horticulture, cropping, livestock farming, or forestry. This population potentially covered smaller farming units than the current census and survey. The survey population in the 1999 survey was AgriQuality New Zealand's national database, AgriBase. Agribase covered all units recorded as holding livestock and/or engaging in grain/arable cropping. In 2002, the population changed back to the Business Frame. Note that 2004–17 deer figures are not directly comparable with those from 2002 and 2003 due to an under-count in deer numbers of about 70,000 deer at 30 June 2002, and 50,000 at 30 June 2003. (For more information, see Data Collection Methodology.) Counts are comparable across geographies and time, as they are estimated using the same methodology. The census question regarding the number of animals is simple and has remained unchanged over time. Census years will not contain sampling error. There will be more random statistical error in survey years, as only approximately 30,000 of the 70,000 farmers are surveyed. Stats NZ estimates an undercount of about 70,000 deer at 30 June 2002, and 50,000 at 30 June 2003. New Zealand and regional totals released on InfoShare website have been through a confidentialising process, with values changed to NA where there are confidentiality risks to respondents.
Changes to time series	
References	Stats NZ (2017). Agricultural production statistics: June 2017 (final). Retrieved 17 January 2019 from www.stats.govt.nz. Stats NZ (nd). InfoShare.